Corrections to "DASK-Net: A Lightweight Dual-Attention Selective Kernel Network for Efficient Dense Prediction in Remote Sensing Imagery"

Furkat Sultonov, Sangseok Yun, *Member, IEEE*, and Jae-Mo Kang, *Member, IEEE*

In the above article [1], the following corrections should be noted:

1) In Section II-A on Page 3: The sentence "Gong et al. [22] implemented dense dilated convolutions with a visual geometry group (VGG) encoder and channel attention to handle occlusions and scale variations." should be corrected to "Gong et al. [22] implemented dense dilated convolutions with a VGG encoder and channel attention to handle occlusions and scale variations."
2) In Section II-B on Page 3: The sentence "Yuan [32] introduced a deep Lightweight global road feature extraction network (CNN) integrating multi layer activations with a signed distance function for robust boundary preservation." should be corrected to "Yuan [32] introduced a deep convolutional neural network (CNN) integrating multi layer activations with a signed distance function for robust boundary preservation."
3) In Section II-C on Page 3: The sentence "Split-attention network based on ResNet (residual network) architecture (ResNeSt) [49] and sample consistency network (SCNet) [50] refine these concepts through feature partitioning and integrated spatial attention." should be corrected to "Split-attention network (ResNeSt) [49] and sample consistency network (SCNet) [50] refine these concepts through feature partitioning and integrated spatial attention."

Additionally, in [1, Tables II–IV], the citation for some baseline models was wrong. Tables II–IV with the correct citations are presented herein.

TABLE II
QUANTITATIVE RESULTS ON THE MASSACHUSETS ROADS DATASET

Methods	Backbone	Params(M)	Size(MB)	MACs(G)	IoU	Precision	Recall	F1 Score
large								
SGCNNet [55]	ResNet50	42.73	170.92	30.05	63.96	79.86	76.26	78.02
DeepLabv3 [47]	ResNet50	39.64	151.43	164.14	63.02	76.63	78.02	77.32
MSMDFFNet [56]	ResNet34	39.26	149.88	149.602	**66.16**	**80.71**	**78.59**	**79.64**
U-Net [57]	-	31.04	118.04	218.95	65.87	80.49	78.38	79.42
LinkNet [58]	ResNet18	**11.53**	**44.04**	**12.08**	64.38	80.26	76.5	78.33
medium								
U-Net++ [59]	MobileOne-S0	9.88	38.06	45.36	65.4	**80.9**	77.34	79.08
UNetFormer [60]	LSKNet-T	7.34	28.58	9.43	63.78	79.03	76.77	77.88
U-Net++ [59]	EfficientNet-lite0	5.93	22.8	22.19	65.2	80.08	77.82	78.94
MACUNet [61]	-	5.15	19.67	33.59	**65.44**	79.79	**78.45**	**79.11**
SegFormer [62]	MiT-B0	3.71	14.17	6.74	60.79	75.22	76.02	75.62
U-Net [57]	MobileNetV3-Small 1.0	3.59	13.73	10.22	63.14	78.43	76.41	77.41
U-Net [57]	MobileNetV3-Small 0.75	2.88	11.04	9.82	63.79	78.85	76.95	77.89
ERFNet [63]	-	2.06	7.9	14.73	65.01	79.82	77.8	78.79
DeepLabV3+ [64]	MobileNetV3-Small 0.75	1.62	6.23	2.62	57.2	73.47	72.09	72.78
MFNET [65]	-	1.34	5.12	4.60	64.33	79.4	77.22	78.29
FastSCNN [66]	-	**1.14**	**4.42**	**0.88**	52.33	68.62	68.8	68.71
small								
ContextNet [67]	-	0.87	3.42	0.89	53.11	67.72	71.13	69.38
DABNet [68]	-	0.75	2.9	5.28	63.88	76.57	**79.41**	77.96
EDANet [69]	-	0.68	2.61	4.45	55.84	70.7	72.64	71.66
SGCPNET [76]	MobileNetV3-Small 1.0	0.61	2.37	0.44	59.79	73.37	76.36	74.84
CFPNet [70]	-	0.55	2.12	4.00	62.95	76.05	78.52	77.26
CGNet [71]	-	0.49	1.92	3.55	63.67	76.44	79.21	77.8
ENet [72]	-	0.36	1.39	2.84	64.69	79.8	77.36	78.56
FSSNet [73]	-	0.17	0.68	1.37	64.22	78.66	77.77	78.21
FPENet [74]	-	**0.11**	**0.46**	**0.78**	60.66	76.27	74.76	75.51
DASK-Net(ours)	-	0.48	1.91	4.85	**65.47**	**80.85**	77.49	**79.13**

Received 3 June 2025; accepted 3 June 2025. Date of current version 18 June 2025. This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean Government [Ministry of Science and ICT (MSIT)] under Grant RS-2025-00559998. *(Corresponding authors: Sangseok Yun; Jae-Mo Kang.)*

Furkat Sultonov and Jae-Mo Kang are with the Department of Artificial Intelligence, Kyungpook National University, Daegu 41566, South Korea (e-mail: furkatsultonov@knu.ac.kr; jmkang@knu.ac.kr).

Sangseok Yun is with the Department of Information and Communications Engineering, Pukyong National University, Busan 48513, South Korea (e-mail: ssyun@pknu.ac.kr).

Digital Object Identifier 10.1109/TGRS.2025.3576881

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

TABLE III
QUANTITATIVE RESULTS ON THE DEEPGLOBE DATASET

Methods	Backbone	Params(M)	Size(MB)	MACs(G)	IoU	Precision	Recall	F1 Score
large								
DeepLabv3 [47]	ResNet50	39.64	151.43	164.14	63.02	76.63	78.02	77.32
MSMDFFNet [56]	ResNet34	39.26	149.88	149.602	**67.67**	**81.55**	**79.91**	**80.72**
U-Net [57]	-	31.04	118.04	218.95	65.28	80.92	77.16	78.99
LinkNet [58]	ResNet18	**11.53**	**44.04**	**12.08**	64.04	79.21	76.98	78.08
medium								
U-Net++ [59]	MobileOne-S0	9.88	38.06	45.36	64.92	**79.51**	77.97	78.73
UNetFormer [60]	LSKNet-T	7.34	28.58	9.43	**65.38**	79.21	**78.92**	**79.07**
U-Net++ [59]	EfficientNet-lite0	6.7	25.77	22.80	64.87	78.72	78.67	78.69
SegFormer [62]	MiT-B0	3.71	14.17	6.74	61.83	75.27	77.59	76.41
U-Net [57]	MobileNetV3-Small 1.0	3.59	13.73	10.22	63.4	78.59	76.64	77.6
U-Net [57]	MobileNetV3-Small 0.75	2.88	11.04	9.82	63.53	78.83	76.6	77.7
DeepLabV3+ [64]	MobileNetV3-Small 0.75	1.62	6.23	2.62	61.87	77.53	75.39	76.44
MFNET [65]	-	1.34	5.12	4.60	64.26	78.9	77.59	78.24
FastSCNN [66]	-	**1.14**	**4.42**	**0.88**	61.92	76.9	76.07	76.49
small								
ContextNet [67]	-	0.87	3.42	0.89	59.77	75.88	73.79	74.82
DABNet [68]	-	0.75	2.9	5.28	64.24	78.34	78.11	78.23
EDANet [69]	-	0.68	2.61	4.45	64.62	78.06	**78.97**	78.51
SGCPNET [76]	MobileNetV3-Small 1.0	0.61	2.37	0.44	62.83	77.75	76.6	77.17
CFPNet [70]	-	0.55	2.12	4.00	63.99	77.78	78.31	78.04
CGNet [71]	-	0.49	1.92	3.55	64.63	78.36	78.67	78.51
ENet [72]	-	0.36	1.39	2.84	63.86	77.08	78.82	77.94
FSSNet [73]	-	0.17	0.68	1.37	60.81	75.29	75.98	75.63
FPENet [74]	-	**0.11**	**0.46**	**0.78**	62.02	77.02	76.1	76.56
DASK-Net(ours)	-	0.48	1.91	4.85	**65.54**	**80.07**	78.31	**79.18**

TABLE IV
QUANTITATIVE RESULTS ON THE WHU BUILDING DATASET

Methods	Backbone	Params(M)	Size(MB)	MACs(G)	IoU	Precision	Recall	F1 Score
large								
BOMSC-Net* [80]	-	129.32	517.28	-	90.15	95.14	94.5	94.8
LFEMAP-Net* [81]	ConvNeXt-B	127.37	509.48	-	**91.09**	**95.81**	94.86	**95.34**
DSAT-Net [40]	ConvNext-Tiny	48.50	202.12	57.60	86.24	91.11	94.16	92.61
BuildFormer [15]	-	40.52	154.99	116.58	90.58	95.17	**94.95**	95.06
DeepLabv3 [47]	ResNet50	39.64	151.43	164.14	85.69	91.73	92.86	92.29
U-Net [57]	-	31.04	118.04	218.95	87.79	93.02	93.99	93.5
UANet [39]	ResNet50	26.73	102.19	29.81	89.45	94.54	94.32	94.43
LinkNet [58]	ResNet18	**11.53**	**44.04**	**12.08**	84.02	89.99	92.69	91.32
medium								
U-Net++ [59]	MobileOne-S0	9.88	38.06	45.36	86.6	94.07	92.72	93.39
UNetFormer [60]	LSKNet-T	7.34	28.58	9.43	89.16	94.77	93.77	94.27
LRAD-Net* [19]	Reg-Net 600	7.3	29.2	6.68	88.89	94.21	93.74	93.86
U-Net++ [59]	EffecientNet-lite0	5.93	22.8	22.19	88.64	94.03	93.93	93.98
DeepLabv3+ [64]	MobileNetv2	5.22	20.05	30.95	**89.2**	94.19	94.45	**94.29**
BiSeNetv2 [77]	-	5.19	19.87	17.79	87.51	93.43	93.25	93.34
EasyNet* [42]	RegNet08	3.90	15.6	3.00	89.16	93.48	**94.94**	94.2
SegFormer [62]	MiT-B0	3.71	14.17	6.74	87.28	93.43	92.99	93.21
U-Net [57]	MobileNetv3-Small 1.0	3.59	13.73	10.24	86.76	93.06	92.76	92.91
RSR-Net* [17]	-	2.9	11.6	3.01	88.32	**94.92**	92.63	-
U-Net [57]	MobileNetv3-Small 0.75	2.88	11.04	9.82	87.31	93.51	92.94	93.22
ERFNet [63]	-	2.06	7.9	14.73	87.58	93.18	93.58	93.38
MFNet [65]	-	1.34	5.12	4.60	89.04	94.44	93.97	94.2
FastSCNN [66]	-	**1.14**	**4.42**	**0.88**	85.34	92.61	91.57	92.09
small								
LETNet [78]	-	0.95	3.71	7.3	88.16	94	93.41	93.71
ContextNet [67]	-	0.87	3.42	0.89	83.84	91.48	90.94	91.22
DABNet [68]	-	0.75	2.9	5.28	87.64	93.18	93.65	93.41
EDANet [69]	-	0.68	2.61	4.45	87.5	93.09	93.57	93.33
SGCPNet [76]	MobileNetv3-Small 1.0	0.61	2.37	0.44	84.69	92.4	91.03	91.71
CFPNet [70]	-	0.55	2.12	4.00	87.01	93.37	92.74	93.05
CGNet [71]	-	0.49	1.92	3.55	88.22	94.16	93.32	93.74
ENet [72]	-	0.36	1.39	2.84	87.09	93.04	93.16	93.1
FSSNet [73]	-	0.17	0.68	1.37	84.64	91.81	91.56	91.68
FPENet [74]	-	**0.11**	**0.46**	**0.78**	84.45	91.81	91.34	91.57
DASK-Net(ours)	-	0.48	1.91	4.85	**89.54**	**94.76**	**94.21**	**94.48**

REFERENCES

[1] F. Sultonov, S. Yun, and J.-M. Kang, "DASK-net: A lightweight dual-attention selective kernel network for efficient dense prediction in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5623416, doi: 10.1109/TGRS.2025.3566097.